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Install and run network diagnostic tools such as TCP dump or wireshark. Start capturing packets on an achieve interface. Open a browser and type the address of your favourite search engine, wait the page load and stop capture, List out the type and the no of every kind of packets captured.

Introduction:

- wireshark is a type of open source packet analyser
- It is used for nlw troubleshooting, analysis, software and communications development.

Why is wireshark refused to as a nlw packet analyer? This is because, it will try to capture nlw packets and try to illustrate that packet data as details as possible.

Purpose:

- Nlw administrator use it to troubleshoot network problems.
- Educational purpose for ~~raising~~ the nlw protocols.

Download and install the wireshark software:

Go to http://www.wireshark.org/download.html and download and install the wireshark binary for your computer.

Teacher's Signature

```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : REDDY-PC
   Primary Dns Suffix . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Unknown adapter McAfee VPN:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-97-07-73-1F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 7 BE200 320MHz #2
   Physical Address. . . . . . . . . : FE-B3-AA-DD-00-1A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 7 BE200 320MHz #3
   Physical Address. . . . . . . . . : FC-B3-AA-DD-00-1B
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%23(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.5.10 (Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, July 24, 2026 9:38:02 nanda
   Lease Expires . . . . . . . . . . : Friday, July 24, 2026 10:09:38 nanda
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.55.1
   DHCPv6 IAID . . . . . . . . . . . : 7885419718
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-D2-E6-93-00-E0-4C-68-CC-02
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:7f4f:3ece%9(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.243.10 (Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Friday, July 24, 2026 9:38:05 nanda
   Lease Expires . . . . . . . . . . : Friday, July 24, 2026 10:09:38 nanda
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.243.254
   DHCPv6 IAID . . . . . . . . . . . : 805326931
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-D2-E6-93-00-E0-4C-68-CC-02
   Primary WINS Server . . . . . . . : 192.168.243.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : gitam.edu
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 7 BE200 320MHz
   Physical Address. . . . . . . . . : FC-B3-AA-DD-00-1A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::d1dc:7fe7:15ce:79c3%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.22.200.120 (Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
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✓) Running Wireshark:

- When you run the wireshark program, you'll get a startup screen, as show below:

- On the upper left of the screen, you'll see the "Interface list", which shows an nlw interface on your computer. Choose an interface to start capturing packets on it.

- To begin capturing packets, click on an interface. Wireshark will show a screen with details of the captured packets.

- To stop capturing, go to the capture menu and select stop.

- File Menu: Save, open, or exit captured data.
  Capture menu: Start (or) stop capturing packets.

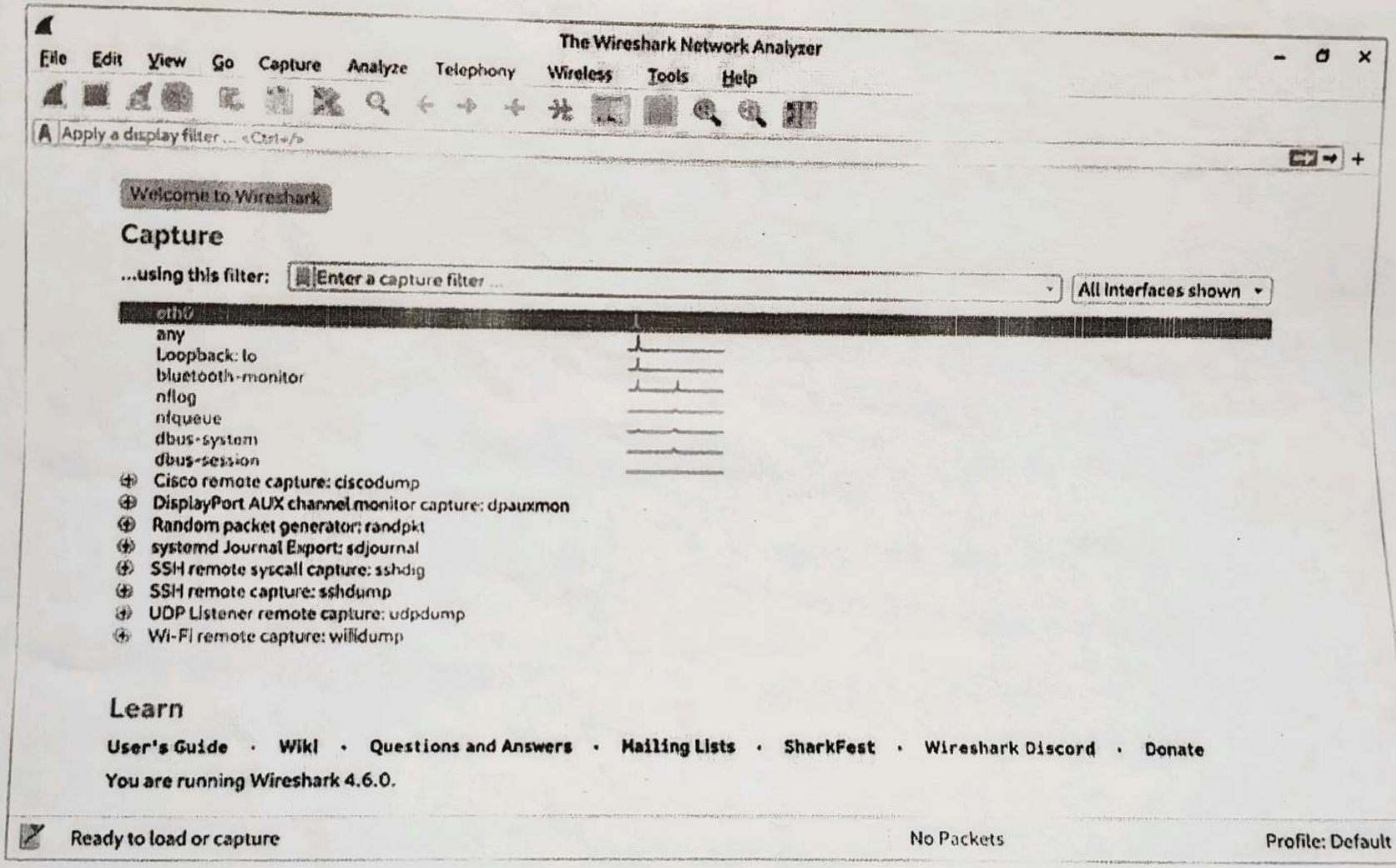
~~1st~~ => Menus:

- File menus: Save, open, or exit captured data
- Captured Menu: Start (or) stop packet capture.

Packet List:

- Shows a summary of each packets (time, source, dest, protocol etc.)
- click a column name to sort the list

Teacher's Signature _____


The Wireshark Network Analyzer
File Edit View Go Capture Analyze Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
Capture
...using this filter: Enter a capture filter ...
All interfaces shown
eth0
any
Loopback: lo
bluetooth-monitor
nflog
nfqueue
dbus-system
dbus-session
Cisco remote capture: ciscodump
DisplayPort AUX channel monitor capture: dpauxmon
Random packet generator: randpkt
systemd Journal Export: sdjournal
SSH remote syscall capture: sshdig
SSH remote capture: sshdump
UDP Listener remote capture: udpdump
Wi-Fi remote capture: wifidump
Learn
User's Guide · Wiki · Questions and Answers · Mailing Lists · SharkFest · Wireshark Discord · Donate
You are running Wireshark 4.6.0.
Ready to load or capture
No Packets
Profile: Default

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Packet details:

- Click a packet to the detailed info, like ethernet IP, TCP/UDP and protocol data.
- Use the plus/minus icons to expand (or) collapse sections.

Fast Filter field:

Enter a protocol name to show only specific packets.

Taking wireshark for a test run:

Assumption: The computer is connected to the internet via a weird Ethernet interface. Indeed, I recommend doing ethernet connection rather than just a wireless connection.

Open Browser:

Launch your web browser to display your homepage

Start Wireshark:

Open wireshark. The main window will appear but it won't capture packets yet

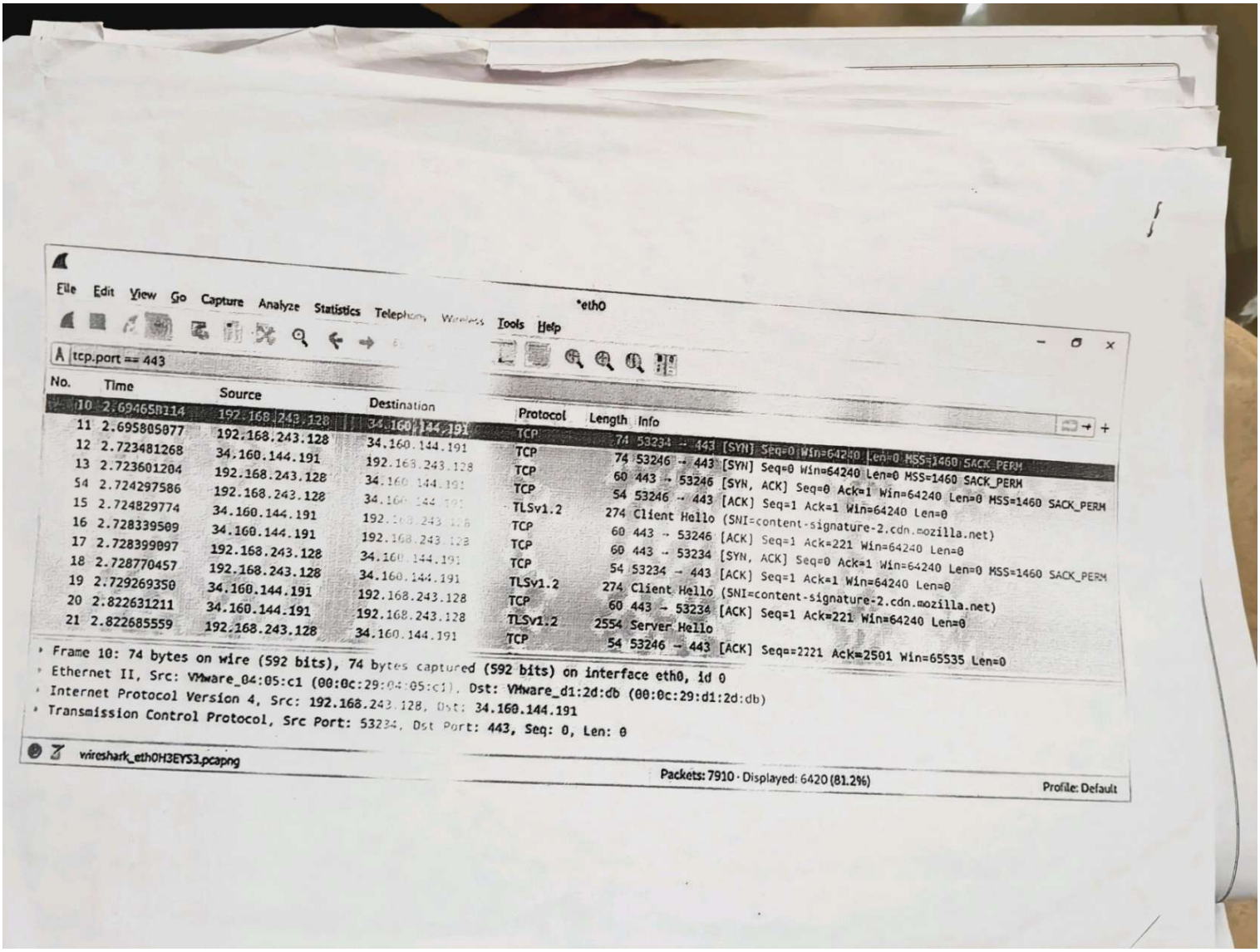
Begin packet capture:

Go to the capture menu, select interfaces, and choose the interface for your nic.

Generate Nlw traffic:

Keep wireshark running & use your browser to visit a website. This creates traffic using HTTP protocol.

Teacher's Signature _____


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
*eth0
tcp.port == 443
No. Time Source Destination Protocol Length Info
10 2.69465B114 192.168.243.128 34.160.144.191 TCP 74 53234 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11 2.695805077 192.168.243.128 34.160.144.191 TCP 74 53246 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
12 2.723481268 34.160.144.191 192.168.243.128 TCP 60 443 → 53246 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
13 2.723601204 192.168.243.128 34.160.144.191 TCP 54 53246 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
54 2.724297586 192.168.243.128 34.160.144.191 TLSv1.2 274 Client Hello (SNI=content-signature-2.cdn.mozilla.net)
15 2.724829774 34.160.144.191 192.168.243.128 TCP 60 443 → 53246 [ACK] Seq=1 Ack=221 Win=64240 Len=0
16 2.728339509 34.160.144.191 192.168.243.128 TCP 60 443 → 53234 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM
17 2.728399097 192.168.243.128 34.160.144.191 TCP 54 53234 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
18 2.728770457 192.168.243.128 34.160.144.191 TLSv1.2 274 Client Hello (SNI=content-signature-2.cdn.mozilla.net)
19 2.729269350 34.160.144.191 192.168.243.128 TCP 60 443 → 53234 [ACK] Seq=1 Ack=221 Win=64240 Len=0
20 2.822631211 34.160.144.191 192.168.243.128 TLSv1.2 2554 Server Hello
21 2.822685559 192.168.243.128 34.160.144.191 TCP 54 53246 → 443 [ACK] Seq=221 Ack=2501 Win=65535 Len=0
Frame 10: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth0, id 0
Ethernet II, Src: VMware_04:05:c1 (00:0c:29:04:05:c1), Dst: VMware_d1:2d:db (00:0c:29:d1:2d:db)
Internet Protocol Version 4, Src: 192.168.243.128, Dst: 34.160.144.191
Transmission Control Protocol, Src Port: 53234, Dst Port: 443, Seq: 0, Len: 0
wireshark_eth0H3EYS3.pcapng
Packets: 7910 · Displayed: 6420 (81.2%)
Profile: Default

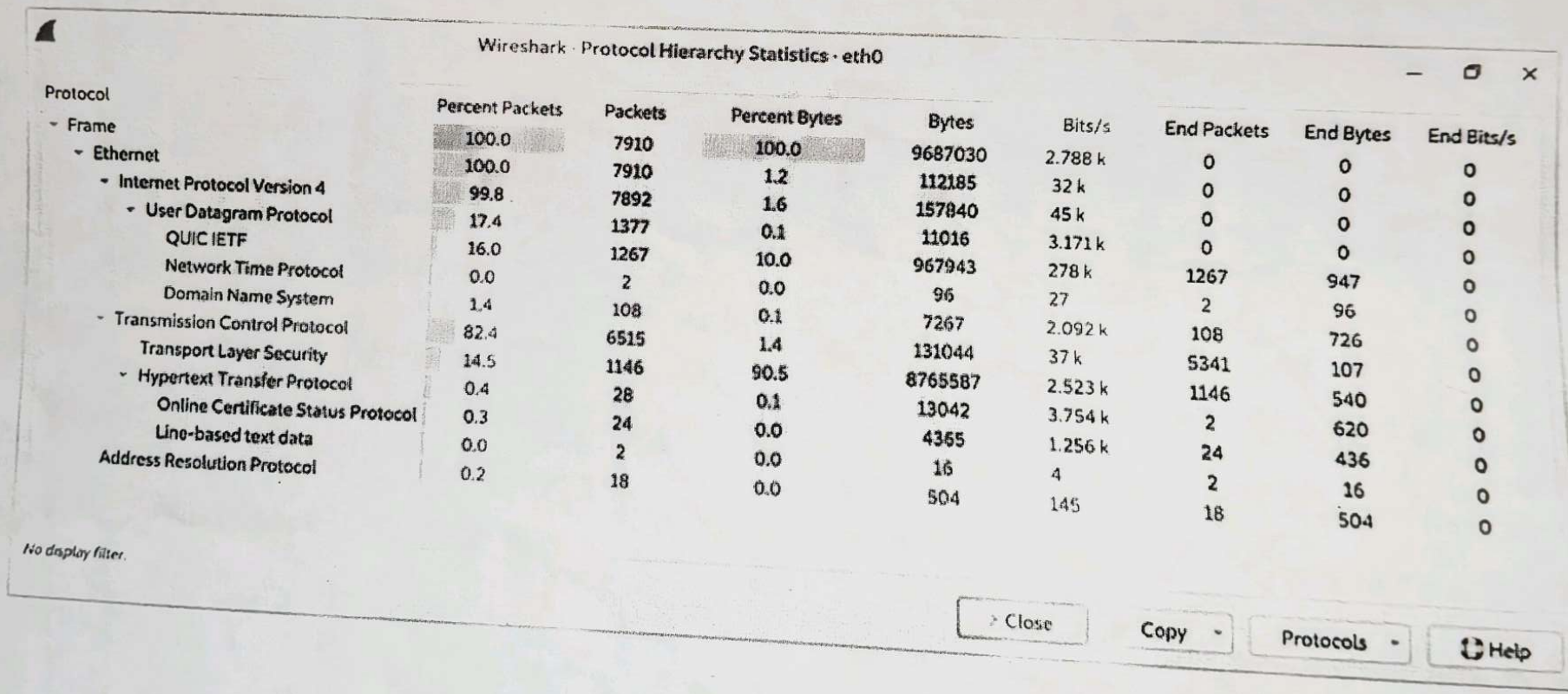
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Stop capturing packets:
After browsing, stop capturing by selecting stop from the capture menu
Filter for HTTP Traffic:
In the display filter field, type http and click Apply to show only HTTP packets.
Find the HTTP GET message:
Look for a packet with GET and the website URL you visited
Select it to view details in the packet header window (Ethernet, IP, TCP & HTTP layer).
Exit Wireshark: Close wireshark when done.

23/01/2026

Teacher's Signature


Wireshark · Protocol Hierarchy Statistics · eth0
Protocol
Frame
Ethernet
Internet Protocol Version 4
User Datagram Protocol
QUIC IETF
Network Time Protocol
Domain Name System
Transmission Control Protocol
Transport Layer Security
Hypertext Transfer Protocol
Online Certificate Status Protocol
Line-based text data
Address Resolution Protocol
Percent Packets
100.0
100.0
99.8
17.4
16.0
0.0
1.4
82.4
14.5
0.4
0.3
0.0
0.2
Packets
7910
7910
7892
1377
1267
2
108
6515
1146
28
24
2
18
Percent Bytes
100.0
1.2
1.6
0.1
10.0
0.0
0.1
1.4
90.5
0.1
0.0
0.0
0.0
Bytes
9687030
112185
157840
11016
967943
96
7267
131044
8765587
13042
4365
16
504
Bits/s
2.788 k
32 k
45 k
3.171 k
278 k
27
2.092 k
37 k
2.523 k
3.754 k
1.256 k
4
145
End Packets
0
0
0
0
1267
2
108
5341
1146
2
24
2
18
End Bytes
0
0
0
0
947
96
726
107
540
620
436
16
504
End Bits/s
0
0
0
0
0
0
0
0
0
0
0
0
0
No display filter.
Close
Copy
Protocols
Help